



YOUNG FERMATS · PRE-ALGEBRA

## CLASS 14

# One-Step Equations

## Multiplication and Division

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### Student Workbook

8 illustrated lessons · 40 practice questions

8 homework review questions · 12 homework word problems

# 1. Understanding Multiplication and Division Equations

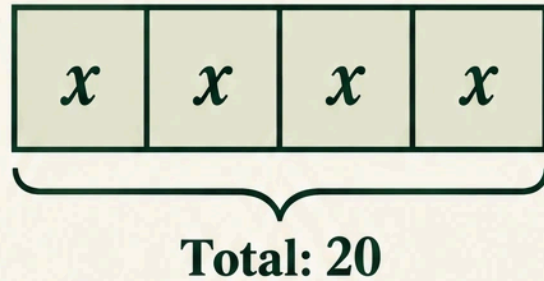
CLASS 14 • One-Step Equations

## Multiplication equation

$$4x = 20$$

$4x$  means  $4 \times x$ .

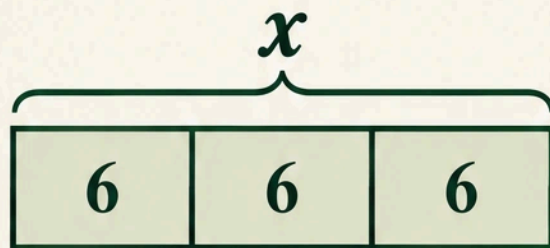
The variable is multiplied by 4.



## Division equation

$$\frac{x}{3} = 6$$

The variable is divided by 3.



**An equation** states that two expressions are equal.

Identify the operation applied to the variable.

## Image 1 practice

Understanding multiplication and division equations

**Q1.** In  $6n = 42$ , what operation is applied to  $n$ ? Find  $n$ .

**Q2.** Read  $\frac{t}{8} = 9$  in words, then solve.

**Q3.** In  $11p = 55$ , identify the coefficient and find  $p$ .

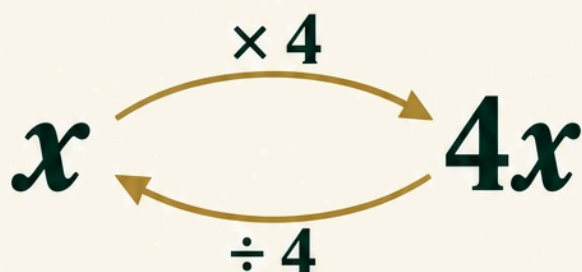
**Q4.** Solve  $12 = \frac{q}{7}$ . Does having the variable on the right change the method?

**Q5.** For  $3x = 24$  and  $\frac{y}{3} = 24$ , predict which unknown is larger, then solve both.

## 2. Using Inverse Operations

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**Multiplication and division undo each other.**



**Undo multiplication  
with division**

$$4x = 20$$

$$\frac{4x}{4} = \frac{20}{4}$$

$$x = 5$$

Divide BOTH sides by 4.

**Undo division  
with multiplication**

$$\frac{x}{3} = 6$$

$$3 \times \frac{x}{3} = 3 \times 6$$

$$x = 18$$

Multiply BOTH sides by 3.

**Keep the equation balanced.**

Apply the same operation to both sides.



## Image 2 practice

Using inverse operations and keeping balance

**Q6.** Complete both blanks with the same number, then solve:  $\frac{7x}{\square} = \frac{63}{\square}$ .

**Q7.** What operation on both sides isolates  $m$  in  $\frac{m}{9} = 5$ ? Solve.

**Q8.** A student changes  $6k = 54$  into  $k = 48$  by subtracting 6. Explain the error and solve correctly.

**Q9.** Show the same operation on both sides to solve  $18 = \frac{n}{4}$ .

**Q10.** A student divides both sides of  $8r = 72$  by 2 and gets  $4r = 36$ . Is that valid? Is  $r$  isolated? Finish solving.

# 3. Solving Multiplication Equations

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**Divide both sides by the coefficient.**

The **coefficient** is the number multiplying the variable.

**Example**

Coefficient: 5

$$5x = 35$$

$$\frac{5x}{5} = \frac{35}{5}$$

$$x = 7$$

Total: 35



Five equal groups total 35.  
Each group is 7.

**Another example**

$$8y = 48$$

$$\frac{8y}{8} = \frac{48}{8}$$

$$y = 6$$

**Key concept.**

Divide BOTH sides by the same nonzero number.

**Check:**

$$5 \times 7 = 35 \quad \checkmark$$

## Image 3 practice

Solving multiplication equations

**Q11.** Solve  $3x = 27$  and check.

**Q12.** Solve  $7a = 84$  and check.

**Q13.** Solve  $96 = 12b$  and check.

**Q14.** Solve  $14t = 119$  and check.

**Q15.** Solve  $18p = 15$  and check.

# 4. Solving Division Equations

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**Multiply both sides by the divisor.**

The **divisor** is the number dividing the variable.

**Example**

Divisor: 4

$$\frac{x}{4} = 6$$

$$4 \times \frac{x}{4} = 4 \times 6$$

$$x = 24$$

Total: 24



Four equal groups of 6 make 24.

**Another example**

$$\frac{y}{5} = 7$$

$$5 \times \frac{y}{5} = 5 \times 7$$

$$y = 35$$

**Key concept.**

Multiply BOTH sides by the same number.

**Check:**

$$24 \div 4 = 6 \checkmark$$

## Image 4 practice

Solving division equations

**Q16.** Solve  $\frac{x}{6} = 9$  and check.

**Q17.** Solve  $\frac{y}{11} = 8$  and check.

**Q18.** Solve  $13 = \frac{z}{5}$  and check.

**Q19.** Solve  $\frac{t}{8} = 2.75$  and check.

**Q20.** Solve  $\frac{p}{12} = \frac{5}{6}$  and check.

# 5. Solving Equations with Negative Numbers

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Use inverse operations and follow the sign rules.

**Multiplication & division:** | Same signs → positive | Different signs → negative

**Negative coefficient**

$$-3x = 18$$

$$\frac{(-3x)}{(-3)} = \frac{18}{(-3)}$$

$$x = -6$$

Divide BOTH sides by  $-3$ .

**Check:** |  $-3 \times (-6) = 18$  ✓

**Negative divisor**

$$\frac{y}{-4} = 5$$

$$-4 \times \left(\frac{y}{-4}\right) = -4 \times 5$$

$$y = -20$$

Multiply BOTH sides by  $-4$ .

**Check:** |  $-20 \div (-4) = 5$  ✓

**Remember the signs.**

$$(+) \times (+) = (+) \quad | \quad (-) \times (-) = (+) \quad | \quad (+) \times (-) = (-) \quad | \quad (-) \times (+) = (-)$$

The same sign rules apply to division.

**Key concept.**

Keep the negative sign attached to its number.

## Image 5 practice

Solving equations with negative numbers

**Q21.** Solve  $-p = -13$  and explain the hidden coefficient.

**Q22.** Solve  $-4x = 28$  and check.

**Q23.** Solve  $9y = -63$  and check.

**Q24.** Solve  $\frac{z}{-6} = -8$  and check.

**Q25.** Solve  $-12q = -9$  and check.

## 6. Solving Equations with Fractions and Decimals

### CLASS 14 • One-Step Equations

Use inverse operations with fractions and decimals.

**Multiplication & division:** | Same operation → undo | Use reciprocals or divide

#### Fraction coefficient

$$\frac{1}{2}x = 8$$

$$2 \times \left(\frac{1}{2}x\right) = 2 \times 8$$

$$x = 16$$

Multiply **BOTH** sides by 2.  
2 is the reciprocal of  $\frac{1}{2}$ .

$$x = 16$$



#### Decimal coefficient

$$0.4x = 2$$

$$\frac{0.4x}{0.4} = \frac{2}{0.4}$$

$$x = 5$$

Divide **BOTH** sides by 0.4.

**Check:** |  $0.4 \times 5 = 2$  ✓

#### Dividing by a fraction

$$\frac{x}{\frac{1}{2}} = 10 \quad x = 10 \times \frac{1}{2} \quad x = 5$$

Multiply both sides by the divisor,  $\frac{1}{2}$ .

#### Key concept.

A fraction coefficient?  
Multiply by its reciprocal.

A decimal coefficient?  
Divide by that coefficient.

## Image 6 practice

Solving equations with fractions and decimals

**Q26.** Solve  $0.3x = 2.7$  and check.

**Q27.** Solve  $\frac{2}{3}y = 10$  using the reciprocal.

**Q28.** Solve  $\frac{z}{0.4} = 7.5$  and check.

**Q29.** Solve  $\frac{5}{8}t = -\frac{15}{4}$  and check.

**Q30.** Solve  $p \div \frac{3}{5} = -\frac{7}{6}$ . Explain which number you multiply by.

# 7. Checking Solutions

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**A solution makes the original equation true.**

## 1. Substitute the value

$$3x = 21$$
$$\text{Try } x = 7$$
$$3 \times 7 = 21 \quad \checkmark$$

Replace  $x$  with 7.  
Both sides equal 21.

## 2. Check a division equation

$$\frac{x}{4} = 6$$
$$\text{Try } x = 24$$
$$24 \div 4 = 6 \quad \checkmark$$

Replace  $x$  with 24.  
Both sides equal 6.

## 3. Catch a mistake

$$5x = 30$$
$$\text{Does } x = 25 \text{ work?}$$
$$5 \times 25 = 125$$
$$125 \neq 30 \quad \times$$

25 does not work.  
Divide 30 by 5 instead.

$$x = 6 \rightarrow 5 \times 6 = 30 \quad \checkmark$$

**Check in the original equation every time.**

## Image 7 practice

Checking solutions and catching errors

**Q31.** Does  $x = 6$  solve  $7x = 42$ ? Show the substitution.

**Q32.** Does  $y = 36$  solve  $\frac{y}{9} = 4$ ? Show the substitution.

**Q33.** A student says  $z = 7$  solves  $-5z = 35$ . Check the claim and correct it.

**Q34.** Which solves  $0.6t = 4.2$ :  $t = 7$  or  $t = 0.7$ ? Check both.

**Q35.** For  $p \div \frac{2}{3} = -\frac{9}{4}$ , Ana gets  $p = -\frac{3}{2}$  and Ben gets  $p = -\frac{27}{8}$ . Who is correct? Verify both values.

## 8. Writing and Solving Word Problems

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Define a variable. Write an equation. Solve and check.

### 1. Example 1 • Cost per notebook

Four equally priced notebooks cost \$28. How much does each notebook cost?

Let  $n$  = the price of one notebook in dollars.

$$\begin{aligned} 4n &= 28 \\ \frac{4n}{4} &= \frac{28}{4} \\ n &= 7 \end{aligned}$$



Each notebook costs \$7.

Check:  $4 \times \$7 = \$28$  ✓

### 2. Example 2 • Equal sharing

A collection of tickets is shared equally among 5 students. Each student receives 6 tickets. How many tickets are in the collection?

Let  $t$  = the total number of tickets.

$$\begin{aligned} \frac{t}{5} &= 6 \\ 5 \times \frac{t}{5} &= 5 \times 6 \\ t &= 30 \end{aligned}$$



There are 30 tickets in total.

Check:  $30 \div 5 = 6$  ✓

Include units and check that your answer makes sense.

## Image 8 practice

Writing and solving word problems

**Q36.** Six equal packs contain 54 stickers altogether. How many stickers are in each pack? Define a variable and write an equation.

**Q37.** A ribbon is cut into seven equal pieces. Each piece is 4 m long. Find the original length using a division equation.

**Q38.** Five equally priced bookmarks cost \$13.75 in total. What does one cost? Write and solve an equation.

**Q39.** A freezer's temperature changes by the same amount each hour. Over three hours, its total change is  $-15^{\circ}\text{C}$ . What is the change per hour?

**Q40.** A tank is three-fourths full and contains 18 liters. What is its full capacity? Define a variable and solve.

## Homework review

R1–R5

**R1.** Image 1: Write and solve an equation for: eight times a number is 56.

**R2.** Image 2: Solve  $\frac{y}{12} = 7$  and name the operation applied to both sides.

**R3.** Image 3: Solve  $16x = 104$  and check.

**R4.** Image 4: Solve  $\frac{x}{15} = \frac{2}{5}$  and check.

**R5.** Image 5: Solve  $-7x = -84$  and check.

## Homework review

R6–R8

**R6.** Image 6: Solve  $0.35x = 2.8$  and check.

**R7.** Image 7: A student claims  $x = 6$  solves  $x \div \frac{3}{4} = 8$ . Is the claim correct? Show a check.

**R8.** Image 8: Two-fifths of the students in a club are 12 students. How many students are in the whole club? Write and solve an equation.

## Homework word problems

W1–W5

**W1.** Five equally priced museum tickets cost \$45. What is the price of one ticket? Write and solve an equation.

**W2.** A supply of markers is divided equally among nine boxes. Each box holds eight markers. How many markers are there in total? Use a division equation.

**W3.** A rectangular banner has an area of 84 square feet and a length of 12 feet. Find its width using a one-step equation.

**W4.** A square garden has a perimeter of 52 meters. How long is each side? Write and solve an equation.

**W5.** Eight identical bottles hold 10 liters altogether. How much does each bottle hold? Write an equation and give a decimal answer.

## Homework word problems

W6–W10

**W6.** A cyclist has completed three-fourths of a route, covering 21 miles. What is the full route length?

**W7.** A game score changes by the same amount in six rounds. The total change is  $-42$  points. What is the change per round? Include its sign.

**W8.** A ribbon is cut into 14 pieces, each 0.75 m long, with no waste. Write a division equation for its original length and solve.

**W9.** A student has read 0.15 of a book, which is 27 pages. How many pages are in the whole book? Write and solve an equation.

**W10.** A model shelf is two-thirds the length of a real shelf. The model is  $3\frac{1}{2}$  feet long. What is the real shelf's length? Write an equation and give an exact answer.

## Homework word problems

W11–W12

**W11.** Identical bags each weigh 1.25 kg. Together they weigh 17.5 kg. Leo subtracts 1.25 from 17.5 to find the number of bags. Explain his error, then write and solve the correct equation.

**W12.** An underwater robot descends at a steady rate of  $\frac{4}{5}$  meter per minute for 7.5 minutes. Let  $d$  be its signed vertical change, with downward changes negative. Write a division equation using the rate, solve, and state both the signed change and the distance descended.